Literature Status for arXiv:1808.00859: Pelczynski Space and GCCR

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Status

This is a `literature_already_answered` packet for a scoped subquestion in Garcia-Lirola–Prochazka, *Pelczynski space is isomorphic to the Lipschitz free space over a compact set*, arXiv:1808.00859. The supporting paper is Hajek–Medina, *Compact retractions and Schauder decompositions in Banach spaces*, arXiv:2106.00331.

Source Question

After proving that there is a compact convex set $K \subset \mathbb{P}$ such that the Pelczynski space $\mathbb{P}$ is isomorphic to $\mathcal{F}(K)$, arXiv:1808.00859 observes that the proof could alternatively be completed by showing that this K is a Lipschitz retract of $\mathbb{P}$. The authors state that they do not know whether this is true, and then recall the broader Godefroy–Ozawa question asking whether every separable Banach space admits a compact convex generating Lipschitz retract.

Supporting Answer

The later paper arXiv:2106.00331 studies exactly this compact-retraction problem. Its introduction states that every Banach space with a finite dimensional decomposition admits a generating convex compact Lipschitz retract. It further states that if the space has a monotone Schauder basis, then such a retract can be chosen with Lipschitz constant $1+\varepsilon$ for every $\varepsilon > 0$.

In Section "Compact Lipschitz retracts", the authors explicitly identify the connection with arXiv:1808.00859: their FDD construction gives a positive answer to the question asked in their citation `GP19`, namely whether Pelczynski space has a GCCR. Since Pelczynski space has a Schauder basis, it falls under the FDD/Schauder-basis theorem.

Scope

The answer is scoped. It shows that Pelczynski space admits some generating convex compact Lipschitz retract. It does not prove that the particular compact set $K = \overline{\mathrm{conv}}\{e_n/n : n \in \mathbb{N}\}$ used in arXiv:1808.00859 is itself a Lipschitz retract. It also does not solve the full Godefroy–Ozawa problem for arbitrary separable Banach spaces, nor the separate compact-model question asking whether every separable infinite-dimensional X admits a compact metric K with $\mathcal{F}(X) \simeq \mathcal{F}(K)$.

Search Evidence

The deterministic lane queue selected arXiv:1808.00859. Cheap run indexes were searched for the arXiv id, title variants, Pelczynski-space keywords, and Lipschitz-free compact-model phrases. No exact packet for arXiv:1808.00859 was found. A local full-source search for compact convex generating Lipschitz retracts found arXiv:2106.00331, which explicitly cites the source paper as `GP19` and states the Pelczynski-space GCCR consequence. Existing run memory already contains a broader no-promotion attempt on the general Godefroy–Ozawa problem, so this packet does not repackage that global open problem as solved.

References

[1] L. C. Garcia-Lirola and A. Prochazka, *Pelczynski space is isomorphic to the Lipschitz free space over a compact set*, arXiv:1808.00859.

[2] P. Hajek and R. Medina, *Compact retractions and Schauder decompositions in Banach spaces*, arXiv:2106.00331.